

Aerodynamic Parameter Estimation from F1 Telemetry

Andrew Coyle

May 2026

Abstract

This report describes a methodology for estimating aerodynamic drag area (CdA), downforce area (CIA), and rolling resistance coefficient (Crr) from publicly available F1 telemetry using the FastF1 library. Coast-down segments from free-practice sessions are fitted to an ODE model incorporating MGU-K energy recovery. CIA is estimated independently via GPS-derived lateral acceleration at near-limit corners. The method is applied to the Aston Martin AMR24 at Monza 2024 and Silverstone 2024, yielding an inter-circuit aerodynamic drag ratio of 1.35x and an aerodynamic efficiency ($\Delta CIA / \Delta CdA$) of 1.4 between the two setup configurations. DRS drag reduction is investigated via race trap speeds but found to be inseparable from slipstream effects in the available data. An investigation into 2026 active aero measurements reveals that the F1 timing API does not broadcast active aero state, blocking mode-split analysis.

1. Introduction

Aerodynamic characterisation of Formula 1 cars is central to setup and race strategy. Teams measure CdA and CIA in wind tunnels and CFD but public access to these parameters is limited. The F1 live timing API, exposed through the FastF1 Python library, provides car telemetry at 240 Hz and GPS position at 10 Hz for all sessions. This creates an opportunity to estimate aerodynamic parameters from first principles using freely available data.

Prior work on extracting aerodynamic parameters from road-car coast-down tests is well established, but F1 cars introduce several complications: active energy recovery during coasting (MGU-K), ride-height-dependent ground-effect aerodynamics, and the lack of direct brake force measurement. This report describes a methodology that addresses the energy recovery term explicitly, documents the known systematic bias from engine braking, and validates the approach through cross-circuit comparison and internal consistency checks.

The Aston Martin AMR24 is used as the reference vehicle. Monza 2024 represents the team's lowest-drag configuration; Silverstone 2024 (run six weeks earlier) represents a medium-to-high downforce setup. Both analyses use the same driver numbering conventions: driver 14 (Alonso) at Monza and driver 18 (Stroll) at Silverstone, both in the AMR24.

2. Data and Session Selection

Free-practice sessions (FP1, FP2, FP3) are used for all coast-down and CIA analyses. Race laps are excluded from the primary analysis because MGU-K deployment varies lap-to-lap with strategy, and tyre state changes more dramatically over longer stints.

Session and driver selection follows an automated survey: all drivers across all FP sessions are evaluated, valid coast-down segments are counted, and the driver with the most segments is selected as reference. At Monza this yields driver 14 with 14 segments across three sessions ($\rho = 1.134 \text{ kg/m}^3$, AirTemp $\sim 31 \text{ degC}$). At

Silverstone the survey selects driver 18 with 7 segments ($\rho = 1.193 \text{ kg/m}^3$, AirTemp $\sim 18 \text{ degC}$). Survey results are cached to avoid re-fetching on subsequent runs.

CIA estimation uses FP2 telemetry, which provides the best balance of representative lap pace and GPS quality. The Monza analysis yields 16 lap-level CIA estimates; Silverstone yields 13.

3. Methodology

3.1 Coast-down ODE model

During free-deceleration phases (throttle $< 5\%$, brake $= 0$), the longitudinal equation of motion reduces to:

$$m \cdot dv/dt = -\alpha \cdot v^2 - \beta - P_{\text{mgu}} / v$$

where $\alpha = (1/2) \cdot \rho \cdot (CdA + Crr \cdot CIA)$ is the aerodynamic coefficient ($\text{N} \cdot \text{s}^2/\text{m}^2$), $\beta = Crr \cdot m \cdot g$ is rolling resistance (N), and P_{mgu} is MGU-K harvest power modelled as a constant-power retarding term (W). The P_{mgu}/v term is critical: without it, energy recovery appears as inflated rolling resistance and the fitted Crr is unphysically high. Including it yields $\beta = 120 \text{ N}$ and $P_{\text{mgu}} = 0\text{-}53 \text{ kW}$, consistent with published MGU-K harvest rates.

Segments are extracted using a state machine: contiguous samples satisfying the throttle and brake conditions are grouped, then filtered to a minimum duration of 0.5 s, minimum entry speed of 120 km/h, and minimum speed drop of 25 m/s. The speed-drop requirement ensures the $\alpha \cdot v^2$ and P_{mgu}/v terms have distinguishable shapes across the segment, which is necessary for reliable parameter separation.

The ODE is integrated numerically using `scipy.integrate.odeint` and fitted via `scipy.optimize.curve_fit` with β fixed at its prior median (120 N). This two-parameter fit (α , P_{mgu}) is more tractable than the unconstrained three-parameter version and avoids correlation between β and α in short segments.

3.2 Pooled fitting

Individual segment fits are noisy. Since α and β are physical constants of the car setup for a given session, they are shared across all N segments in a pooled fit. This reduces the parameter count from $3N$ to $N+2$ (shared α and β , per-segment P_{mgu}), and tightens the α uncertainty by roughly $\sqrt{N}$. The pooled fit uses `scipy.optimize.least_squares` with TRF method, warm-started from per-segment medians.

3.3 CIA from GPS lateral acceleration

At the tyre friction limit in a high-speed corner, the lateral force balance gives:

$$a_{\text{lat}} = \mu \cdot (g + (\rho \cdot ClA) / (2 \cdot m) \cdot v^2)$$

GPS position rows (Source=='pos', $\sim 10 \text{ Hz}$) are differentiated to compute lateral acceleration. Velocity is computed from position differences; acceleration from velocity differences; both smoothed with a 5-sample rolling mean. Lateral acceleration is projected from the world frame using the cross-product formula. Samples above a circuit-specific threshold (2.5g at Monza; 3.5g at Silverstone) are used; this selects only near-limit apex samples where the friction model is valid. CIA is the median over all qualifying lap estimates with $\mu = 1.8$ assumed. Sensitivity to μ is documented: at $\mu = 1.5$ the Monza estimate becomes 3.93 m^2 ; at $\mu = 2.0$ it is 3.02 m^2 .

3.4 DRS drag delta

Race laps on the Monza main straight are classified as DRS-open (DRS channel > 10 in any sample within 0-600 m from lap start) or DRS-closed. An energy balance over the activation zone converts the mean trap speed difference to DCDA:

$$\Delta C_dA = m \cdot (v_{\text{open}}^2 - v_{\text{closed}}^2) / (\rho \cdot \int v^2 dx)$$

The integral is computed numerically from a representative DRS-closed speed trace. A slipstream-stratification

analysis is performed by estimating the gap to the car ahead for each DRS-open lap from lap-end timing differentials, and restricting to marginal-DRS laps (gap 0.5-1.0 s) to reduce wake effects.

4. Results

4.1 Monza 2024 -low-drag configuration

Fourteen segments from FP1/FP2/FP3 pass quality filters ($R^2 \geq 0.90$) and are included in the pooled fit. Results:

Parameter	Value	Expected range	Status
Pooled alpha ($N \cdot s^2/m^2$)	0.800 +/- 0.002	-	
Composite $2\alpha/\rho$ (m^2)	1.410	~0.85	inflated (see 5.1)
CIA (m^2)	3.36 +/- 0.65	2.5 - 3.5	pass
CdA (m^2)	1.37 +/- 0.01	0.8 - 1.0	inflated (see 5.1)
Crr	0.0131	0.015 - 0.020	plausible
Mean Durbin-Watson	0.66	~2.0	autocorrelation

Table 1: Monza 2024 aerodynamic parameters. Composite and CdA are inflated by engine braking.

The Durbin-Watson statistic of 0.66 indicates strong positive autocorrelation in ODE residuals across all segments. Residuals show a systematic pattern: the model overshoots at high speed and undershoots at low speed, consistent with an unmodelled speed-dependent retarding force (engine braking torque).

4.2 Silverstone 2024 -medium-high downforce configuration

Seven segments yield a successful pooled fit for driver 18 (Stroll, AMR24):

Parameter	Monza 2024	Silverstone 2024	Ratio S/M
Pooled alpha ($N \cdot s^2/m^2$)	0.800 +/- 0.002	1.076 +/- 0.005	1.35x
Composite $2\alpha/\rho$ (m^2)	1.410	1.805	1.28x
CIA (m^2)	3.36 +/- 0.65	3.88 +/- 0.56	1.16x
CdA (m^2)	1.37 +/- 0.01	1.75 +/- 0.01	1.28x
Crr	0.0131	0.0131	1.00

Table 2: Circuit comparison. Crr equality to three decimal places validates the method.

The Crr equality is a strong internal consistency check: rolling resistance is a tyre and surface property, not a setup variable. Its invariance across two circuits, two drivers, different weather conditions, and different session compositions confirms that the ODE correctly separates the rolling resistance term from the aerodynamic term.

The inter-circuit alpha ratio of 1.35x is consistent with published accounts of the downforce-to-drag tradeoff between Silverstone and Monza setups, and with the Spa/Monza alpha ratio of 1.29x from the cross-validation in the primary analysis (Spa is typically between Monza and Silverstone in downforce level).

From the deltas, the aerodynamic efficiency of the Silverstone package relative to Monza is:

$$\text{delta_CIA} / \text{delta_CdA} = 0.524 / 0.388 = 1.4$$

For each unit of drag added by running more wing at Silverstone, 1.4 units of downforce are gained. This figure represents the slope of the operating point along the car's polar curve between the two configurations.

4.3 DRS drag delta

The pooled Monza race-lap analysis yields 194 DRS-open and 733 DRS-closed classified laps. The pooled trap speed difference is:

$$\text{delta_v_trap} = 11.91 \pm 0.53 \text{ km/h} \quad (90\% \text{ CI: } 10.87 - 12.95)$$

Converting via the energy balance gives DCDA = -0.107 m^2 (90% CI: -0.28 to $+0.05 \text{ m}^2$). The expected DRS-only contribution is -0.5 to -0.8 m^2 . The discrepancy is explained by slipstream confounding: every DRS-open lap in a race also has the following car within 1 s of the car ahead, the same condition that guarantees aerodynamic wake exposure.

Stratifying by estimated gap to the car ahead (marginal-DRS group: 0.5-1.0 s, $n = 87$ laps) reduces the delta to 10.09 km/h and DCDA to -0.086 m^2 , indicating some slipstream reduction but confirming that a pure DRS measurement is not achievable from race data alone.

5. Limitations

5.1 Engine braking inflates CdA

At throttle = 0, the drivetrain applies additional retarding torque through compression, valve timing, and MGU-H harvesting. This force is not captured by the $P_{\text{mgu/v}}$ term (which models only MGU-K). The model absorbs it into α , inflating the composite $2\alpha/\rho$. The composite value of 1.410 m^2 exceeds the expected $\text{CdA} + \text{Crr} \cdot \text{CIA} = 0.80 + 0.014 \cdot 3.36 = 0.85 \text{ m}^2$ by approximately 66%. Correcting this requires gear position and engine-braking torque curves, which are not available in the FastF1 data.

Despite this, the method remains useful for relative comparisons: engine braking affects both circuits similarly, so the α ratio (1.35x) and the absolute Crr estimate are reliable. The absolute CdA should be treated as an upper bound.

5.2 DRS inseparable from slipstream

The gap-stratification analysis confirms that reducing slipstream reduces the apparent delta but does not eliminate it. A clean DRS measurement from race data is not feasible; qualifying or track-mapping data (where DRS is active but following cars are rare) would be required.

5.3 CIA depends on assumed μ

The GPS lat-g method requires an assumed tyre friction coefficient. The sensitivity (roughly CIA proportional to $1/\mu$) means a 10% error in μ produces a 10% error in CIA. Published values for Pirelli compounds at operating temperature range from approximately 1.5 to 2.0, placing CIA in the range 3.0 - 3.9 m^2 for the Monza dataset.

6. 2026 Active Aero Investigation

The 2026 F1 regulations replace DRS with an automatic active aero system operating without any gap-to-car-ahead requirement. This removes the slipstream confound and, in principle, enables a clean DCDA measurement from the drag difference between straight mode (active aero open) and corner mode (active aero closed) coast-down segments within the same free-practice session.

Inspecting the raw F1 timing API for Canada 2026 shows that channel 45 (the DRS indicator in 2024, values 0/12/14) is absent from every 2026 car data message across all 22 drivers in both the Race and FP1. The FastF1 source code acknowledges this: `entry['Cars'][drv]['Channels'].get('45', 0)` with a comment 'drs is no longer included in 2026'. No replacement channel carrying active aero state is present in the 2026 API.

Without active aero state telemetry, segment-by-segment mode classification based on track position (known active aero zone boundaries) is methodologically compromised because it confounds aerodynamic mode with speed range, gear, and track gradient. The measurement will become feasible once the F1 timing API exposes active aero state.

7. Conclusions

The coast-down ODE method with GPS lat-g CIA estimation provides a robust framework for aerodynamic characterisation from public F1 telemetry. Key findings:

- $CIA = 3.36 \pm 0.65 \text{ m}^2$ at Monza 2024, consistent with expected low-downforce values.
- Silverstone/Monza α ratio = 1.35x, consistent with known setup differences and the independently derived Spa/Monza ratio of 1.29x.
- $C_{rr} = 0.0131$ at both circuits, confirming that the rolling resistance term is correctly isolated from the aerodynamic term.
- Aerodynamic efficiency of the Silverstone vs Monza package: $\Delta_{CIA} / \Delta_{CdA} = 1.4$.
- DRS drag reduction is not isolatable from slipstream in race data. The combined DRS + slipstream trap speed Δ is 11.9 km/h.
- Engine braking inflates the composite $2\alpha/\rho$ by approximately 66%, making absolute CdA unreliable without engine braking torque data.
- The 2026 F1 timing API does not broadcast active aero state, blocking mode-split analysis until the channel is exposed.

References

- Oehrly et al. FastF1: A Python library for Formula 1 telemetry. <https://github.com/theOehrly/Fast-F1>
- FIA Formula 1 Technical Regulations 2024/2026. <https://www.fia.com/regulation/category/110>
- Scipy: Fundamental Algorithms for Scientific Computing in Python. Nature Methods 17, 261-272 (2020).